Sumit Kumar

480-740-7728 sumitdvlp@gmail.com linkedin.com/sumitdvlp/ github.com/sumitdvlp

Experience

Uber Aug 2022 – Present
Software Engineer *Seattle, Washington*

Optimally matching riders with drivers at Airport using low-latency platform powering millions of real-time match decisions per second and improving marketplace health at airports of global scale using ML powered systems.

Skills: Distributed Systems, Time series forecasting, Event Driven architecture, Microservices, Experimentation.

Airport Supply Team - Project Eng Lead (Medium to XL size projects) *Apr 2024 – Present*

- L size - Built a centralized Rule Engine platform to make low latency decisions for every promotion Airport originated trip the platform is handling 10k RPS supporting $<=$ 200ms latency requirement.
- L size - Engineered and built a suite of ML-powered forecasting features — including Estimated-to-Request (in-queue and off-queue), Earnings per Hour, and proactive driver notifications—improving marketplace reliability and driving a +3% uplift in Complete/Request, total 1.5% increase in airport GB growth, using time-series forecasting.
- M size - Designed, developed a reliable automatic exit/entry platform for drivers, partnered with Airport authorities. The product helped reduce frictions with partners and reduce airport congestion and operational cost.
- XL size - Designed, Engineered and launched the On-Demand driver feature, improving driver availability during peak airport demand, delivering a +2.5% Complete/Request uplift, +0.2% Gross Bookings increase, and 5% reduction in rider wait time, deployed across all major global airports.
- Lead pods of 4+ BE, Coordinated with Product, Mobile, Design, City Operations, and Matching teams to deliver real-time airport driver experiences, enabling large-scale model deployment, real-time inference.

Airport Matching Team *Aug 2022 – Apr 2024*

- Introduced offer broadcasting for airport drivers, integrating a new solver and an ML-based acceptance prediction model to increase airport queue throughput by 8–10%.
- Extended offers to off-airport drivers during moderate low-supply hours, reducing rider wait times by 10–15% across major and minor airports.
- Improved ETA calculations for riders and drivers to enhance objective-function accuracy, reducing pickup wait times and lowering post-match cancellations by 2%.

Microsoft Azure MySql/Postgres Db migration platform team Mar 2019 – Aug 2022
Software Engineer *Redmond, Washington*

Azure provides a free service enabling enterprises to migrate on-prem, AWS RDS dbs to Azure Data Platform.

- Built tools, services, and applications to help scenarios for automatic migration with zero downtime.
- Developed an online PostgreSQL migration solution using WAL logs, native replication, and dump-restore; the product migrated 5k medium size databases per year to Azure Data platforms.
- Founding engineer for the offline migration pipeline; delivered an offline MySQL migration service with a 3× faster median migration time for typical databases. Designed the online solution using reactive programming and native DB features (binlog, replication).

Open Source , Side projects Oct 2021 – Present
Code contribution and recommendations *Redmond, Washington*

- Contributed to the MySqlConnector open-source driver, adding support for multiple data types and proposing improvements to telemetry, connection statistics, and writer efficiency.
- Proposed changes issues related to telemetry, connection stats, and writer efficiency, and efficiency changes.
- Implemented a PyTorch-style a autograd engine from scratch and validated its gradients against the original library.

Technical Skills

Programming Languages & Frameworks: GoLang, Java, Python, C#, Pl/SQL, .NET, Shell(Bash), Git, Reactive fx, Django
Tools & Technologies: gRPC, Apache Thrift, Protobuf, RESTful, Pytorch, Grafana, Apache Spark, Replication.
Machine Learning: Transformers, Time series forecasting, Regression, Feature Selection, Deep Neural Networks
Storage Databases: Apache Kafka, Postgres, Redis, MySQL, Azure Data, NoSql databases, AWS S3.

Education

Arizona State University 2018-2019
Masters of Science in Computer Science *Tempe, Arizona*